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Cam-actuated self-locking clamp assembly for solar panels

Abstract

Clamp assemblies and systems for mounting solar panels to mounting units are presented. For instance, a clamp assembly for clamping a solar panel includes a clamp portion and a lever portion. The clamp portion includes at least one tooth for clamping onto a first flange of the solar panel. The lever portion is spaced apart from the clamp portion and include a cam and a guard. The cam includes a rounded surface and a flat surface. Each of the clamp portion and the lever portion comprise an opening for receiving a pivoting element. The opening is generally aligned when the clamp portion and the lever portion are fitted together. Upon rotation of the lever portion, the cam urges the clamp portion into a clamped condition.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application claims priority to U.S. Provisional Patent Application Ser. No. 63/284,792, filed on Dec. 1, 2021, the entirety of which is incorporated herein for all purposes.

BRIEF SUMMARY

(1) In one embodiment, a clamp assembly for clamping a solar panel includes a clamp portion and a lever portion. The clamp portion includes at least one tooth for clamping onto a first flange of the solar panel. The lever portion is spaced apart from the clamp portion and include a cam and a guard. The cam includes a rounded surface and a flat surface. Each of the clamp portion and the lever portion comprise an opening for receiving a pivoting element. The opening is generally aligned when the clamp portion and the lever portion are fitted together. Upon rotation of the lever portion from a position in which the rounded surface of the cam rotates along the bottom of a support portion to a position in which the flat surface of the cam locks against the bottom of the support portion, the cam urges the clamp portion into a clamped condition such that the at least one tooth clamps onto the first flange of the solar panel.

(2) In another embodiment, a clamp assembly for mounting a solar panel to a mounting unit is presented. For example, the clamp assembly includes a saddle portion, a clamp portion and a lever portion. The saddle portion has a means for mounting the saddle portion to a tower of the mounting unit. The clamp portion is fitted into a top of the saddle portion and comprising at least one tooth for clamping onto a first flange of the solar panel. The lever portion is fitted into a bottom of the

saddle portion. The lever portion comprises a cam and, in one example, a guard. The cam comprises a rounded surface and a flat surface. Each of the saddle portion, the clamp portion and the lever portion comprise an opening for receiving a pivoting element. The openings are generally aligned when the saddle portion, the clamp portion and the lever portion are fitted together. Upon rotation of the lever portion from a position in which the rounded surface of the cam rotates along the bottom of the saddle portion to a position in which the flat surface of the cam locks against the bottom of the saddle portion, the cam urges the clamp portion into a clamped condition such that the at least one tooth clamps onto the first flange of the solar panel. In an example, upon continued rotation of the lever portion the guard of the lever portion contacts the bottom of the saddle portion to prevent further rotation of the lever portion.

(3) In a further embodiment, a system for mounting a solar panel to a plurality of mounting units is presented. For example, a plurality of clamp assemblies as described in the first embodiment discussed above are positioned on towers of the mounting units and are used to clamp a solar panel in place at an angle to the mounting units.

(4) An advantage realized in one or more embodiments described herein is that toolless installation may be achieved because the clamps may be attached to the solar panel using the cam-actuated self-locking clamp assembly instead of using a tool. Another advantage realized in one or more embodiments described herein is that the clamp can be first attached to the solar panel and then the solar panel can be attached to the mounting units, creating great efficiencies in the installation of solar panels. Another advantage realized in one or more embodiments described herein is that the clamping assembly is universal to solar panels by virtue of clamping a bottom flange of the solar panel, and thereby being able to clamp any thickness solar panel.

(5) The above embodiments are exemplary only. Other embodiments as described herein are within the scope of the disclosed subject matter.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

(1) So that the manner in which the features of the disclosure can be understood, a detailed description may be had by reference to certain embodiments, some of which are illustrated in the accompanying drawings. It is to be noted, however, that the drawings illustrate only certain embodiments and are therefore not to be considered limiting of its scope, for the scope of the disclosed subject matter encompasses other embodiments as well. The drawings are not necessarily to scale, emphasis generally being placed upon illustrating the features of certain embodiments. In the drawings, like numerals are used to indicate like parts throughout the various views, in which:

(2) FIG. 1A illustrates a saddle portion of a clamp assembly, in accordance with one or more aspects set forth herein; and

(3) FIG. 1B illustrates a clamp portion of the clamp assembly, in accordance with one or more aspects set forth herein; and

(4) FIG. 1C illustrates a lever portion of the clamp assembly, in accordance with one or more aspects set forth herein; and

(5) FIGS. 1D-1I illustrate the partially assembled clamp assembly, in accordance with one or more aspects set forth herein;

(6) FIG. 2A illustrates two mounting units with clamp assemblies shown on posts of the mounting units, in accordance with one or more aspects set forth herein;

(7) FIGS. 2B & 2C illustrate a solar module assembled to the clamp assemblies on the posts of the two mounting units of FIG. 2A, in accordance with one or more aspects set forth herein;

(8) FIGS. 3A-3D illustrate examples of the operation of the clamp assemblies, in accordance with one or more aspects set forth herein;

(9) FIGS. 4A-4D illustrates an example in which the clamp assembly does not include a saddle portion, and instead the clamp portion clamps to a square tube, in accordance with one or more aspects set forth herein; and

(10) FIGS. 5A-5D illustrates an example in which the clamp assembly does not include a saddle portion, and instead the clamp portion clamps to a square tube, in accordance with one or more aspects set forth herein.

(11) Corresponding reference characters indicate corresponding parts throughout several views. The examples set out herein illustrate several embodiments, but should not be construed as limiting in scope in any manner.

DETAILED DESCRIPTION

(12) The present disclosure relates to the installation of solar panels on a roof of a building or structure. The prior art describes the use of mounting units with towers of different heights that allow a solar panel to be installed at an angle relative to the roof and mounting units. In some conventional systems, rails are used between the towers of the mounting unit, and the solar panel is installed on top of these rails. In such a case, rails are used to mount the solar panels without tools and in the regular spacing required.

(13) For example, U.S. Pat. No. 8,850,754 B2 to the same inventor of the present disclosure, entitled Molded Solar Panel Racking Assembly, was granted Oct. 7, 2014, and is incorporated herein in its entirety for any purpose including for the purpose of enablement and written description. The '754 Patent requires the use of rails extending between the towers of the mounting units in order to support the solar panels. These rails can be expensive to procure and install. Applicant has discovered that the present technique, in some embodiments, can eliminate these expensive rails. Applicant's technique makes use of clamp assemblies described below, and rails are not required.

(14) During installation of solar panels, conventional systems require that a variety of tools and components are carried and used by the installer. For instance, multiple bolts and other fasteners are typically required for installation of each solar panel. This forces the installer to perform hundreds of discrete tasks using tools and hundreds of separate components while stationed on the roof of a building or structure. Clearly this can lead to increased labor costs when installing solar panels, in addition to other disadvantages. In addition, bolts and screws can tend to loosen over time due to vibration from the wind and other sources, meaning that continued maintenance tasks are required to ensure operation of the installation.

(15) FIG. 1A illustrates a saddle portion **110** of a clamp assembly **100** (shown assembled in FIGS. 1D-1F). The saddle portion **110** may be attached to posts of a mounting unit **210** or **220** (shown in FIGS. 2A-2C) using opening Y. Slots **112** facilitate assembly and opening X lines up with the openings X of the other portion when the clamp assembly **100** is assembled, as described below.

(16) FIG. 1B illustrates a clamp portion **120** of the clamp assembly **100** (shown assembled in FIGS. 1D-1F). Clamp portion **120** includes tabs **124** which include openings X, which may be inserted into slots **112** of the saddle portion **110**. Clamp portion **120** includes teeth **122** which will grip the frame of a solar panel, as described below. Clamp portion **120** may also include ribs **125** which add to the structural integrity of claim portion **124**, which will assist in gripping the frame of a solar panel, as described below. For instance, ribs **125** strengthen the clamp portion **120** to mitigate loosening of the at least one tooth clamped onto the first flange of the solar panel when the clamp portion **120** is in the clamped condition, as described below.

(17) FIG. 1C illustrates a lever portion **130** of the clamp assembly **100** (shown assembled in FIGS. 1D-1F). Lever portion **130** includes cams **131** which include openings X which align with the openings X of the saddle portion **110** and clamp portion **120** when assembled together. Lever portion **130** includes a flat portion **136** used to rotate the lever along the axis of openings X, and guards **134** which prevent over-rotation. Cams **131** include flat surfaces **132** and curved surfaces **133**. In the embodiment of FIG. 1C, D1 is the diameter of cam **131** to the curved surface as shown,

and D2 is the diameter of cam **131** to the flat surface **132** as shown, and D2 is greater than D1, facilitating the locking of the solar panel by the clamp as explained below.

(18) FIGS. **1D-1F** illustrate the partially assembled clamp assembly **100**. As may be seen, the saddle portion **110**, clamp portion **120** and lever portion **130** are assembled together such that openings X align. Rivets may be used to join the clamp assembly **100** at openings X. The tolerances of the clamp assembly are such that when the flat portion **136** of the lever portion **130** is pushed downward, the lever portion **130** rotates clockwise such that the flat surfaces **132** unlock from the saddle and the lever portion releases its force on the clamp portion **120** to unlock the clamp portion **120** from the top of the saddle portion **110**. In addition, guard portion **134** prevents further rotation in the counterclockwise direction, preventing over-torqueing and preventing potential damage to a clamped solar panel.

(19) FIG. **1F** depicts the lever portion **120** in the loose, unlocked or unclamped condition. As depicted in FIG. **1F**, lever **136** is rotated to about a .sub.5 o'clock position, such that diameter D1 is oriented upwards, such that lever portion **120** causes clamp portion **120** to be loose such that teeth **122** are not urged against the saddle **110**. FIG. **1G** depicts the lever portion **120** in the locked or clamped condition. As depicted in FIG. **1F**, level **136** is rotated to about a 3 o'clock position, such that diameter D2, which is greater than diameter D1, is oriented upwards, thus urging clamp portion **120** downward such that teeth **122** are disposed against the saddle portion **110**.

(20) FIG. **1H**, like FIG. **1F**, depicts the lever portion **120** in the loose, unlocked or unclamped condition, this time with frame portion **310** of solar panel **300** placed between the clamp portion **120** and the saddle portion **110**. As depicted in FIG. **1H**, lever **136** is rotated to about a 5 o'clock position, such that diameter D1 is oriented upwards, such that lever portion **120** causes clamp portion **120** to be loose such that teeth **122** are not urged towards the saddle **110** and thus teeth **122** do not engage with the frame portion **310** (which may be described as the bottom flange of solar panel). FIG. **1I**, like FIG. **1G**, depicts the lever portion **120** in the locked or clamped condition, this time with frame portion **310** of solar panel **300** placed between the clamp portion **120** and the saddle portion **110**. As depicted in FIG. **1F**, the frame portion is the bottom flange of the solar panel. As depicted in FIG. **1F**, level **136** is rotated to about a 3 o'clock position, such that diameter D2, which is greater than diameter D1, is oriented upwards, thus urging clamp portion **120** downward such that teeth **122** engage frame portion **310** and hold it against the saddle portion **110**.

(21) Advantageously, by clamping the bottom flange of a solar panel, the clamps described herein are more universal in use because, unlike conventional fasteners that fasten the top flange of a solar panel, the thickness of the solar panel need not be accounted for. In conventional fasteners, different part numbers will need to be stocked for different solar panels of different thickness. Of course, conventional fasteners such as bolts or screws cannot be installed on a lower flange, unlike the toolless clamps of the present disclosure. Furthermore, by clamping the bottom flange of the solar panel, it is less likely that the fastening will be loosened because the cam lock is below the plane of the top surface.

(22) FIGS. **2A-2C** illustrates two mounting units **210** and **220** with clamp assemblies **100** shown on posts **201** of the mounting units **210**, **220**. As depicted in FIG. **2B**, a solar panel **300** may be positioned across four clamp assemblies **100** and clamped into place, with the details of the frame **310** being clamped as shown in FIGS. **1F-1I**.

(23) FIGS. **3A-3D** illustrate example embodiments of the operation of the clamp assemblies **100** when used with solar panels **300** and mounting units **210**. For instance, FIG. **3A** depicts a clamp assembly **100** attached to a mounting unit **210**. A frame portion **310** of a solar panel **300** is then clamped by the clamp assembly **100**. As shown in FIGS. **3B-3C**, by spacing a plurality of mounting units **210**, a plurality of solar panels **300** may be clamped using a plurality of clamping units **210**. In addition, FIG. **3A** shows how the post of a basket is received into the saddle and abuts against the clamping mechanism to prevent it from being loosened because the cam actuated clamp cannot be rotated when in this position. In such a manner, the clamping assembly **100** is resistant to

vibration, weather related motions, etc., so that it will not come undone.

(24) FIGS. 4A-4D illustrates an example in which the clamp assembly **100** does not include a saddle portion, and instead the clamp portion clamps to a square tube **400**. FIG. 4A depicts the clamp assembly **100** in the clamped condition and placed over the square tube **400** and ready to clamp a frame portion **310**. FIG. 4B depicts the clamp assembly **100** of FIG. 4A along line 4B-4B thereof, with the frame portion **310** elided. FIG. 4C depicts the clamp assembly **100** in the clamped condition and clamping the frame portion **310** to the square tube **400**. FIG. 4D depicts the clamp assembly **100** of FIG. 4C along line 4D-4D thereof, with the frame portion **310** elided.

(25) FIGS. 5A-5D illustrates an example in which the clamp assembly does not include a saddle portion, and instead the clamp portion clamps to a square tube, in accordance with one or more aspects set forth herein. FIG. 5A depicts the clamp assembly **100** in the clamped condition and placed over the round tube **500** and ready to clamp a frame portion **310**. FIG. 5B depicts the clamp assembly **100** of FIG. 5A along line 5B-5B thereof, with the frame portion **310** elided. FIG. 5C depicts the clamp assembly **100** in the clamped condition and clamping the frame portion **310** to the round tube **500**. FIG. 5D depicts the clamp assembly **100** of FIG. 5C along line 5D-5D thereof, with the frame portion **310** elided. As depicted in FIGS. 4A-5D, modifications to the shape of the clamp assembly **100** to accommodate different cross-sectional tubes, such as square, round, rectangular, etc., are within the scope of the present disclosure.

(26) Thus, the clamp assembly **100** may operate in conjunction with various support portions, including a saddle **110** or tubes **400**, **500** in order to clamp a solar panel to the support portion. In such a manner, the present disclosure is compatible with both railed and railless installation.

Claims

1. A clamp assembly for clamping a solar panel, the clamp assembly comprising: a clamp portion comprising at least one tooth for clamping onto a first flange of the solar panel; a lever portion spaced apart from the clamp portion, the lever portion comprising a cam and a guard, the cam comprising a rounded surface and a flat surface; wherein each of the clamp portion and the lever portion comprise an opening for receiving a pivoting element, the opening being generally aligned when the clamp portion and the lever portion are fitted together; and wherein upon rotation of the lever portion from a position in which the rounded surface of the cam rotates along the bottom of a support portion to a position in which the flat surface of the cam locks against the bottom of the support portion, the cam urges the clamp portion into a clamped condition such that the at least one tooth clamps onto the first flange of the solar panel.
2. The clamp assembly of claim 1, wherein upon continued rotation of the lever portion, the guard of the lever portion contacts the bottom of the support portion to prevent further rotation of the lever portion.
3. The clamp assembly of claim 1, wherein the support portion comprises a tube and the clamp portion and the lever portion are fitted to the tube.
4. The clamp assembly of claim 3, wherein the tube is disposed between the clamp portion and the pivoting element when the clamp assembly is fitted to the tube.
5. The clamp assembly of claim 1, wherein the support portion comprises a saddle portion, the saddle portion including an opening that is generally aligned with the openings of the clamp portion and the lever portion for receiving the pivoting element.
6. The clamp assembly of claim 5, wherein the saddle portion further comprises an additional opening for affixing the saddle portion to a fixed structure.
7. The clamp assembly of claim 1, wherein the first flange of the solar panel is the lower flange of the solar panel, and the clamp assembly is configured to clamp solar panels having variable thicknesses by clamping the lower flange.
8. A clamp assembly for mounting a solar panel to a mounting unit, the clamp assembly

comprising: a saddle portion with a means for mounting the saddle portion to a tower of the mounting unit; a clamp portion fitted into a top of the saddle portion and comprising at least one tooth for clamping onto a first flange of the solar panel; a lever portion fitted into a bottom of the saddle portion, the lever portion comprising a cam and a guard, the cam comprising a rounded surface and a flat surface; wherein each of the saddle portion, the clamp portion and the lever portion comprise an opening for receiving a pivoting element, the opening being generally aligned when the saddle portion, the clamp portion and the lever portion are fitted together; wherein upon rotation of the lever portion from a position in which the rounded surface of the cam rotates along the bottom of the saddle portion to a position in which the flat surface of the cam locks against the bottom of the saddle portion, the cam urges the clamp portion into a clamped condition such that the at least one tooth clamps onto the first flange of the solar panel; and wherein upon continued rotation of the lever portion, the guard of the lever portion contacts the bottom of the saddle portion to prevent further rotation of the lever portion.

9. The clamp assembly of claim 8, wherein the clamp portion further comprises at least one rib disposed on the body of the clamp portion such that the rib strengthens the clamp portion to mitigate loosening of the at least one tooth clamped onto the first flange of the solar panel when the clamp portion is in the clamped condition.

10. The clamp assembly of claim 8, wherein the clamp assembly is riveted at the opening.

11. A system for toolless and railless mounting of a solar panel to a plurality of mounting units using a plurality of clamp assemblies, each clamp assembly of the plurality of clamp assemblies comprising: a saddle portion with a means for mounting the saddle portion to one tower of one mounting unit; a clamp portion fitted into a top of the saddle portion and comprising at least one tooth for clamping onto a flange of the solar panel; a lever portion fitted into a bottom of the saddle portion, the lever portion comprising a cam and a guard, the cam comprising a rounded surface and a flat surface; wherein each of the saddle portion, the clamp portion and the lever portion comprise an opening for receiving a pivoting element, the opening being generally aligned when the saddle portion, the clamp portion and the lever portion are fitted together; wherein upon rotation of the lever portion from a position in which the rounded surface of the cam rotates along the bottom of the saddle portion to a position in which the flat surface of the cam locks against the bottom of the saddle portion, the cam urges the clamp portion into a clamped condition such that the at least one tooth clamps onto a flange of the solar panel; wherein upon continued rotation of the lever portion, the guard of the lever portion contacts the bottom of the saddle portion to prevent further rotation of the lever portion; and wherein each clamp assembly of the plurality of clamping assemblies is mounted to a different tower of the plurality of mounting units such that the solar panel is mounted at an angle to the mounting units.

12. The system of claim 11, wherein the clamp portion further comprises at least one rib disposed on the body of the clamp portion such that the rib strengthens the clamp portion to mitigate loosening of the at least one tooth clamped onto the first flange of the solar panel when the clamp portion is in the clamped condition.

13. The system of claim 11, wherein the clamp assembly is riveted at the opening.

14. The system of claim 11, wherein the solar panel comprises a top flange and a bottom flange, wherein the flange of the solar panel clamped by the clamp portion is the bottom flange of the solar panel, wherein the toolless mounting facilitates the ability to clamp to the bottom flange notwithstanding that the bottom flange is below the top flange of the solar panel.
